Seat No._____

GANPAT UNIVERSITY
M.Sc.(IT) SEM-II EXAMINATION

P42A4DAV: DATA ANALYSIS AND VISUALIZATION

April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) (30)

A. What is MongoDB? Discuss the advantages of MongoDB in brief.

B. Give the difference between SQL vs NoSQL.

C. How to Insert and Delete Document in MongoDB explain with example.

D. Explain the aggregate() method in MongoDB with example.

E. Discuss the MongoDB Database References in detail with example.

F. Explain Auto-Increment Sequence in MongoDB with example.

G. Explain the Heatmaps and Scatter Charts in brief.

H. Explain Line and Area Charts in MongoDB.

2. Answer the following: (ANY ONE) (06)

A. Explain the data types use in MongoDB.

B. What is NoSQL? Discuss the Types of NoSQL databases.

3. Answer the following: (ANY TWO) (08)

A. Define the terms: Document, Collection, Database, ensureIndex()

B. Explain the find(), skip(), pretty() and sort() functions in MongoDB.

C. How to Create and Drop Collection in MongoDB explain with example.

4. Answer the following: (ANY TWO) (08)

A. Explain MapReduce in MongoDB with example.

B. Write a short note on ObjectId in MongoDB,

C. Explain Regular Expression in MongoDB with example.

5. Answer the following: (ANY TWO) (08)

A. Discuss the use of Geospatial Charts in brief.

B. Explain the features and elements of MongoDB Charts in brief.

C. Explain the use of Data Tables and Number Charts in brief.

END OF PAPER

Seat No. _____

GANPAT UNIVERSITY
M.Sc. (IT) SEM-II (CBCS)
P42A3JP2: Java Programming Techniques -II

Time: 3 Hours] April - June 2023 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

1. **Answer the following: (Any Five)** **(30)**

1. What is servlet? Discuss life cycle of servlet.
2. Write a short note on standard actions.
3. Discuss different types of JSF Html tags in detail.
4. Write a JSP program for insert and select operation.
5. Explain architecture of Hibernate.
6. Write a short note on page directives.
7. Discuss types of hibernate annotation.
8. List out the methods used for reading form data. Explain it with example.

2. **Answer the following: (Any One)** **(5)**

1. List out session tracking techniques. Explain any one with example.
2. Discuss RequestDispatcher in detail.

3. **Answer the following: (Any One)** **(10)**

1. What is the use of scripting elements? Explain example.
2. What is JSP? Write down advantages over Servlet.

4. **Answer the following: (Any Two)** **(10)**

1. List out different types of events in JSF. Explain any one in detail.
2. Explain different phases of JSF life cycle.
3. What is navigation? Explain anyone.

5. **Answer the following: (Any One)** **(5)**

1. Write a short note on ORM.
2. Write a hibernate program for insert operation.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.(IT) SEM-II EXAMINATION (CBCS)

April - June 2023

P42A2WD2: Web Designing - II

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX)** **(30)**

A. Explain the MVC architecture of AngularJS with an appropriate example.

B. What is the currency filter in AngularJS ?

C. What is $rootscope and how do we use it?

D. Discuss AngularJS CSS in brief.

E. How many types of data binding are there in AngularJS?

F. Explain the **ng-repeat** directive.

G. Discuss the Events: ng-Click, Mouse with suitable example.

H. Explain the AngularJS expressions with a suitable example.

2. **Answer the following: (ANY TWO)** **(07)**

A. Distinguish between AngularJS and JavaScript expressions.

B. Explain the differences between one-way binding and two-way binding.

C. Discuss AngularJS Array with appropriate example.

3. **Answer the following: (ANY TWO)** **(07)**

A. What is AngularJS Module? Explain how to create a Module?

B. Define the term controller. Explain it with example in brief.

C. Write a short note on: **ng-app**.

4. **Answer the following: (ANY TWO)** **(08)**

A. Explain AngularJS Form Validation with appropriate example.

B. Discuss the Services: **$http, $timeout**.

C. Explain the built-in filters with suitable examples.

5. **Answer the following: (ANY TWO)** **(08)**

A. Discuss the HTML DOM in brief.

B. What is Angular JS animation? Explain step by step how to Animate an HTML element?

C. How can you integrate AngularJS with HTML?

END OF PAPER

Enrollment No

GANPAT UNIVERSITY

M.Sc.(IT) Sem-II EXAMINATION

P42A1UML: UNIFIED MODELLING LANGUAGE

Time: 3 Hours] April - June 2023 [Total Marks: 60

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX)** (30)

a. Explain Collaboration diagram in detail with suitable example.

b. Explain common mechanism used in UML.

c. What is UML? Explain visualizing, specifying, constructing and documenting terms.

d. Explain Generalization Relationship with suitable example.

e. What is forking and joining? explain it with suitable example.

f. Explain Interface, type and role with suitable example.

g. Explain Behavioral things in Basic Building blocks.

h. Explain multiplicity with example.

2. **Answer the following: (ANY ONE)** (06)

a. Which are basic building blocks? Explain Structural things in Detail.

b. Explain Software development lifecycle in detail.

3. **Answer the following: (ANY TWO)** (08)

a. What is association class? Explain with example.

b. What is class? How to give name, attribute and operations with class in detail?

c. Explain "Has-a" kind of relationship with suitable example.

4. **Answer the following: (ANY TWO)** (08)

a. What is Package? How can assign name and owned elements in package.

b. Create a class Diagram for Online examination system.

c. Explain Importing and exporting in package.

5. **Answer the following: (ANY TWO)** (08)

a. What is usecase diagram? Explain it in detail also explain the components of use case diagram

b. Explain Sequence diagram in detail and with example.

c. Explain activity diagram in detail with suitable example.

END OF PAPER

Enrollment No

GANPAT UNIVERSITY

M.Sc.(IT) Sem-II EXAMINATION (CBCS)

P42A3AT2 - Advance Technology - II (.Net)

Time: 3 Hours] [Total Marks: 60

April - June 2023

Instructions:

1. The figures to the right indicate full marks.
2. Be precise and to the point in your answer.

Q-1. Answer the following: (Any Six) (30)

I. Explain the model view controller pattern in detail.

II. Explain ViewBag and ViewData using examples.

III. Explain the conditional statement of the razor.

IV. Explain the different ways to install entity framework core with appropriate commands.

V. Explain the migration command of entity framework with an example.

VI. Explain the area in detail.

VII. Explain the filter pipeline in detail.

VIII. How you can consume WEBAPI using JQuery ajax.

Q-2. Answer the following: (Any Two) (10)

I. Explain the layout and viewimport views with examples.

II. What is routing? Explain types of routing in detail.

III. Explain the return types of an ActionMethod using an example.

Q3. Answer the following: (Any One) (06)

I. Create the employee model with at least five attributes and use any three data annotations.

II. Explain the code first and database first approach in detail.

Q-4. Answer the following: (Any One) (06)

I Write a code for the action filter which is executed before and after the action method of any controller.

II What is a filter? Explain the types of filters in detail.

Q5. Answer the following: (Any Two) (08)

I Explain HTTP verbs in detail.

II What is JSON? Explain in detail with examples.

III Write a code to create WEBAPI that generates any single random number between 1 and 100 and compares it with the parameter passed by the user to the action method. If it is matched then return the message "You win" or else "You loss".

END OF PAPER